Bladder Cancer & Diversion

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 129	Tumours of the Urinary Bladder
Ch 130	Management Strategies for NMIBC
Ch 131	Muscle-Invasive & Metastatic Bladder Cancer
Ch 132	Surgical Management of Bladder Cancer
Ch 133	Intestinal Segments in Urinary Diversion
Ch 134	Continent Urinary Diversion
Ch 135	Minimally Invasive Urinary Diversion
Ch 124	Bladder Surgery for Benign Disease (incl. vesicovaginal fistula)
★	Putting it together

129 Tumours of the Urinary Bladder

1 · Epidemiology (headline)

- ~550,000 cases, 200,000 deaths (2018). **Fourth most prevalent**; **costliest per patient lifetime**.
- Median age **73**; ~9 in 10 after 55. **M:F ≈ 3:1**.
- **Females present later, worse stage for stage**: haematuria treated as urinary tract infection (UTI).

2 · Risk factors

Exposure predicts histology: aromatic amines → urothelial carcinoma (UC); chronic inflammation → squamous/glandular.

- P2 CA bladder, etiology, newer investigations and treatment — *risk-factor table (occupation, arsenic, Lynch/MSH2, obesity, diet)*
- **Smoking**: **30–40%** of UC; risk **2–3×**; pipes, cigars, e-cigarettes too. **N-acetyltransferase 2 (NAT2) slow acetylators** highest risk (acetylation detoxifies amines). Near never-smokers after **>15 years** quit. ~20% still smoke at diagnosis → more non-muscle-invasive (NMIBC) recurrence and muscle-invasive (MIBC) cancer death.
- Occupation also: tobacco and petroleum workers.
- **Radiation**: cervical EBRT 2–4×, prostate 1.5–2×; brachytherapy smaller; IMRT little or none.
- **Genetics**: NAT2/GSTM1 matter with exposure. Germline variants in 13–24% of UC (MSH2, BRCA1/2), high-grade not low-grade NMIBC → germline testing in selected high-grade.
- **Schistosoma haematobium** (only species): eggs in vesical venules → Th2 inflammation, fibrosis → **SCC**; potentiates smoking. Catheter in spinal cord injury, stones → SCC. **Exstrophy**: malignancy 3.3–7.5%, **>90% adenocarcinoma**.
- **Cyclophosphamide** (only proven chemotherapeutic): **phosphoramide mustard** († acrolein → haemorrhagic cystitis, not cancer); ~4.5×, >20 g, latency peaks 10–14 years.
- Diet signals vanish once smoking is controlled; supplements do not protect.

3 · Pathology

Normal urothelium 4–7 cells: basal → intermediate → **umbrella cells** (urine–blood barrier).

Benign lesions (reactive mimics)

Lesion	Hallmark	Potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recur; resect
Urothelial papilloma	Delicate stalks; FGFR3 mutant, TP53/RB wild type	None (formerly low-grade Ta); resect
Nephrogenic adenoma	Tubule-like glands after injury, transplant, intravesical therapy	None; re-resect if recurs
Leukoplakia	Keratinising squamous metaplasia; white flakes	**Not premalignant** (no TP53) → no follow-up

Lesion	Hallmark	Potential / action
Cystitis cystica/glandularis	From von Brunn nests	Glandularis: low adenocarcinoma risk → resect, treat cause, survey

Flat (precursor) lesions

Loss of polarity = dysplasia; loss of umbrella cells = carcinoma in situ (CIS).

Lesion	Key features	Significance
Hyperplasia	≥10 layers, no atypia	Flat may precede low-grade papillary (FGFR3)
Reactive atypia	Proportionate to inflammation; polarity kept	Coexists with dysplasia/CIS → investigate
Atypia of unknown significance	Disproportionate; CK20, CD44	Limited utility
Dysplasia	Loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** → CIS if prior cancer
CIS	Flat, high grade, noncohesive pleomorphic; umbrella cells lost	Precursor of invasive high-grade

UC grade (2004 WHO) – noninvasive spectrum

NMIBC = ~**70%**: Ta, CIS, T1.

	PUNLMP*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chr 9 loss; FGFR3, PI3K, RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20%/5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

~80% have mixed (mostly squamous) differentiation. **Altered growth patterns behave worse than stage**.

Variant	Hallmark	Chemo-sensitive?	Management
Micropapillary (0.7–2.2%)	Clusters with retraction artefact; **any amount counts**; ERBB2, luminal	No	**Early radical cystectomy (RC) even for NMIBC** (BCG fails unless fully resected); MIBC immediate RC
Sarcomatoid (0.3–0.6%)	Large infiltrative mass; same clone as carcinosarcoma	Poor	Immediate RC ± intraoperative radiation
Plasmacytoid (<1%; incl. signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, relapse common	Neoadjuvant chemotherapy (NAC) whenever possible

Variant	Hallmark	Chemo-sensitive?	Management
Nested (M:F 6:1)	Mimics von Brunn nests, inverted papilloma	No	Often upstaged; stage-matched outcome as UC
Squamous (16–22%, up to 60%)	Basal subtype	NAC active in mixed	Recurrence, progression
Glandular (~10%)	Gland-like spaces	–	As conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC (not lung lineage)	2–5% in West; majority where schistosomiasis endemic	**Secondary > primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic (hypercalcaemia, Cushing); TP53, RB1	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; no level 1 data; 5-FU–cisplatin by analogy	**En bloc dome + urachal ligament + umbilicus**

> **KEY POINT**
> Small cell = systemic: primary cystectomy gives poor DFS not rescued by adjuvant chemotherapy; neoadjuvant markedly improves cancer-specific survival. SCC in spinal cord injury fell after switch to intermittent self-catheterisation – inflammation drives it.

Urachal (2016 WHO): dome/anterior, epicentre in wall; no widespread cystitis glandularis; secondary excluded. TNM unsuitable.

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum
IV	Metastatic

4 · Molecular biology

	Papillary pathway	Flat pathway
Origin → product	Hyperplasia → low-grade papillary	Dysplasia → CIS → high-grade invasive
Genetics	**FGFR3** (up to 80% low-grade Ta vs 10–20% MIBC), chr 9 deletion, PIK3CA (25% NMIBC)	**TP53** (up to 76% MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS **mutually exclusive with FGFR3** (same pathway), equal in NMIBC/MIBC. RB1 loss → E2F3 overexpression. PTEN loss + TP53 → metastasis. FGFR3–TACC3 fusions are

oncoproteins.

TCGA subtype	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3, FOXA1
Behaviour	Less aggressive; papillary	Aggressive; squamous; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; checkpoint inhibitors; EGFR

- **Consensus classification (2020), six classes**: luminal papillary (24%; best, median OS 4 y), luminal nonspecified, luminal unstable, stroma-rich, **basal/squamous (35%, commonest)**, neuroendocrine-like (3%; RB1 loss; worst, ~1 y).
- TCGA: luminal-infiltrated – poor cisplatin NAC response (try immunotherapy); neuronal → etoposide–cisplatin; APOBEC-high survives better. MCQ Micropapillary maps to luminal.

5 · Clinical presentation and evaluation

- **Painless gross haematuria in 85%**; microscopic in nearly all. Irritative symptoms suggest CIS.
- Evaluate every gross haematuria, even if later urinalysis negative or on anticoagulants. PSA: ~10% of recurrent gross haematuria is prostate cancer. Bimanual at TURBT if invasive *→ Ch 132*.
- Negative work-up → near-zero risk for 6 years; repeat at 3–5 years if persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	CT urogram (CTU)	CTU (AUA 2012); ultrasound may replace
Cytology/markers	Reasonable	Avoid routinely

Dipstick (haem peroxidase): sensitivity 91–100%, specificity ~65% → **confirm with microscopy**.

Cystoscopy and enhanced imaging

Cystoscopy + biopsy = gold standard; flexible = rigid. White light (WLC) misses small papillary and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate → porphyrins in tumour → red fluorescence under blue light (360–450 nm)	No dye; 415 and 540 nm absorbed by haemoglobin → dark vessels
Benefit	More Ta and CIS; CIS only on BLC in ~¼; office BLC finds ~1/5 recurrences missed by WLC	Better CIS detection; RCT – 12-month recurrence same overall, lower in low-risk
Status	Adjunct; only HAL FDA approved	Adjunct

Urine cytology and biomarkers

- P1 P3 Urine cytology; Asymptomatic haematuria and urinary biomarkers — *good for high grade only; Paris System (HGUC N:C ≥0.7); NMP22, BTA, FISH, ImmunoCyt; none replaces cystoscopy*

> **ALGORITHM**
> Positive high-grade cytology + negative cystoscopy → random bladder and prostatic urethral biopsies → selective upper-tract washings ± ureteroscopy

6 · Staging and risk groups

Stage	TNM, AJCC 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)
I	T1 – lamina propria
II	T2a inner / T2b outer half muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **2017 change**: regional nodes without M1 = stage III (was IV). MCQ Common iliac = N3, not M1a.
- **T1a/T1b** (vs muscularis mucosae) recommended: ~50% of T1 upstaged to ≥T2 at cystectomy; deeper = worse.
- **Prostatic urethra**: mucosal CIS not adverse but escapes intravesical therapy; only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta, multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

1 · Definition and natural history

- **NMIBC**: urothelial carcinoma (UC) not invading detrusor – Ta, T1, carcinoma in situ (CIS). **70–80%** of new bladder cancers: Ta 60–70%, T1 20–30%, CIS ~10% → *Ch 129*.
- Recurrence **40–60%**; progression ~6% (low-grade [LG] Ta) to 17% (high-grade [HG] T1).
- **Grade, not stage, is the strongest predictor of progression**; CIS second. HG Ta and HG T1 progress similarly. Primary CIS progresses in **>50%**; untreated CIS reaches muscle in 40–83%.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in **one-third to one-half**; clinical T1 unreliable.

> **KEY POINT**
> NMIBC is two diseases: LG = **recurrence** problem, HG = **progression** problem. Chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) treats progression, cystectomy is for uncontrollable progression.

2 · Risk stratification

Risk is **dynamic** – reassess at every recurrence.

AUA/SUO 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; PUNLMP	Single post-op chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG + maintenance to **3 yr**. BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene, gemcitabine–docetaxel

EORTC and CUETO scoring models

Factor	EORTC	CUETO
Number, prior recurrence rate, T stage, concurrent CIS, grade (WHO 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC (1-/5-yr): **recurrence** ← prior rate, number, size; **progression** ← T stage, CIS, grade.
- EORTC **overestimates** (little maintenance BCG, perioperative mitomycin C [MMC] or re-TURBT); CUETO derived in BCG-treated patients → fits better. MCQ Both use **WHO 1973**.

3 · Endoscopic management

TURBT: principles

- Diagnostic and therapeutic (technique *→ Ch 132*). Quality levers: **complete resection** and **detrusor in specimen**; missing muscle → higher mortality.
- Image upper tracts at diagnosis (CTU or retrograde): tumours **<5%**; risk: trigonal, CIS, high risk.
- **Diverticular tumour**: no detrusor → beyond lamina propria = **T3a**. LG: cold-cup + fulguration. HG: strongly consider partial cystectomy or RC.
- **En bloc resection**: fewer perforations, more detrusor, better staging; **same recurrence**.

Repeat (second-look) TURBT

- P4 Re-look (second-look) TUR bladder tumour — *indications (all T1, incomplete, no muscle; consider HG Ta), within 6 weeks; residual ≥40%, HG T1 upstaged 25–30% (~50% without muscle); residual T1 → 80% progression*

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Cold-cup biopsies (no cautery artefact). Random biopsies **not** in low risk; consider if positive cytology/high risk. Prostatic urethral biopsy if CIS or bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences: office fulguration/laser – needs **prior LG diagnosis and negative cytology** (no tissue). **Active surveillance** of small LG tumours safe in selected.

Enhanced cystoscopy

	Blue-light (HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins fluoresce red under blue light; intravesical instillation	415 nm (superficial) and 540 nm (deeper) absorbed by haemoglobin; no agent
Gain	~**one-third more CIS**; smaller papillary	More papillary tumours and CIS
Weakness	Cost; pragmatic RCT: no 3-yr recurrence benefit	False positives ~60%

- HAL replaced 5-ALA (no photosensitisation). No proven progression benefit; AUA: use if available.

4 · Single post-operative instillation

- P3 Immediate post TURBT intravesical Therapy for TCC — *anti-implantation; within 6 h (useless at 24 h); MMC, epirubicin, gemcitabine 2 g (SWOG S0337); best in solitary LG; withhold if perforation; never BCG*
- MMC within **6 h**; level 1 evidence; limited gain in recurrent/multiple tumours. Thiotepa ineffective.
- Perforation risks: chemical cystitis, desquamation, contracture, rare deaths.

5 · Intravesical chemotherapy

- P4 Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy — *MMC optimisation (Au 2001); Gem/Doce regimen; device-assisted; BCG mechanism, schedule, toxicity*

- Reduces **recurrence**, **no proven effect on progression**. Intermediate risk: MMC or gemcitabine = BCG, especially LG multifocal recurrent; 6-week induction → monthly for 1 yr.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	FDA-approved for BCG-refractory CIS; CR ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- EMDA/chemohyperthermia (Synergo, HIVEC) beat passive MMC.
- **Gem/Doce** (regimen → P4-068; maintenance in responders): also increasingly BCG-naive HG; phase III vs BCG under way.

6 · Bacille Calmette-Guérin (BCG)

Mechanism

Fibronectin binding → cytokines, immune-cell influx → **Th1** cytotoxicity (→ P4-068). IL-10 drives suppressive Th2. No strain superior; age and immunosuppression blunt response.

Administration

> **ALGORITHM**
> Wait **2–4 weeks** after TURBT (re-epithelialisation) → urinalysis before each dose (exclude infection, gross bleeding) → 1 vial in 50 mL saline by catheter → retain 1–2 h (restrict fluids, diuretics, caffeine before). Traumatic catheterisation → delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	UTI; liver disease (no isoniazid for sepsis); previous TB; anti-TNF drugs; poor performance status; advanced age

- Most absolute = **breached urothelium → intravasation**. *Safe in transplant/steroids, weaker response.

Efficacy

- **Only intravesical agent reducing progression** – only in trials with **maintenance**; beats MMC only with maintenance.
- CIS initial CR up to **84%**; ~half durable; only ~30% free of recurrence/progression at 10 yr.
- Second induction for persistent CIS → further 30–50% CR; beyond 2 courses progression 20–50% (~7%/course). **6-month response predicts prognosis.**

Schedule

ALGORITHM

SWOG (Lamm 2000): induction weekly ×6 → 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**. High risk: full dose 3 yr. Intermediate: 1 yr (EORTC 30962).

Toxicity management

Grade	Features	Management
1	Cystitis, mild haematuria, fever <38.5°C, <48 h	Symptomatic; consider culture
2	Severe or >48 h	Culture, CXR, LFTs; isoniazid + rifampicin + pyridoxine; resume at ½–⅓ dose
3	Haemodynamic change, persistent fever; allergic (arthralgia, rash); organ (liver, lung, kidney)	**Stop BCG**; isoniazid + rifampicin + ethambutol; prednisone only **with** antimycobacterial cover; ID input

- MCQ BCG resistant to **pyrazinamide**. Avoid cycloserine (psychiatric). Asymptomatic granulomatous prostatitis: no treatment.
- Quinolone 24–48 h after each instillation reduces cystitis and delays, no loss of efficacy.

BCG shortage

ALGORITHM

Prioritise high risk → reduce dose (especially maintenance) → split vials → chemotherapy for intermediate (single agent) and high risk (Gem/Doce) → early RC for highest risk.

- TICE is the only FDA-approved strain.

7 · Recurrence after BCG

- P2 BCG failure — *definitions table: adequate BCG (≥5/6 induction + ≥2/3 maintenance; T1 induction alone), unresponsive (HG within 6 months Ta/T1, 12 months CIS), resistant (reinduce, CR 30–50%), intolerant, delayed relapse*

KEY POINT

Trap: HG T1 at 3 months after induction alone = unresponsive. CIS or HG Ta at 3 months = "resistant" → more BCG.

- After failure check prostatic urethra and upper tracts: urethral/upper tract recurrence in ~half on long follow-up.

Bladder-sparing options (unfit/declining RC)

Agent	Mechanism / schedule	CIS CR 3 → 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-PD-1; 200 mg IV 3-weekly up to 2 yr	41% → 19%	63%; grade 3–4 13% (immune)
Nadofaragene firadenovec*	Non-replicating adenovirus with IFN-α2b gene + Syn3; intravesical 3-monthly	53% → 24%	71%; grade 3–4 4%
N-803 + BCG	IL-15 superagonist → CD8+ T and NK cells	71% (any time) → 45%	~85%

Agent	Mechanism / schedule	CIS CR 3 → 12 mo	Cystectomy-free
Gem/Doce	Weekly ×6, monthly ×2 yr	— → 60% (43% at 2 yr)	84%; HG RFS 65%/52% at 1/2 yr; no level 1 evidence

*FDA-approved. CR halves by 12 months; none matches cystectomy. Radiotherapy palliative, no effect on CIS.

8 · Early radical cystectomy in high-risk NMIBC

- T1 failing BCG at 3 months: **82%** progress; CIS ~50%. Prolonged intravesical therapy: ~¼ alive with bladder at 15 yr. Immediate RC for clinical T1: 10-yr DFS **92%**, 64% if understaged. Even 12-week delay worsens survival.
- **Progression to muscle invasion under surveillance does worse than de novo MIBC.**

ALGORITHM

Strongly consider RC: BCG-unresponsive HG · persistent HG T1 on re-resection · T1 with CIS, LVI or variant histology · deep lamina propria invasion · diffuse CIS · diverticular HG tumour · distal ureteric or prostatic ductal involvement · tumour unresectable completely · patient preference.

9 · Surveillance

Cystoscopy + cytology = gold standard. Recurrence best predicted by **tumour at 3-month cystoscopy** and number at first resection.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	3 months → annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly 1–2 yr → 6–12-monthly 2 yr → annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr** → 6-monthly 2 yr → annually **for life**	Same schedule	Annually 2 yr, then lengthen

- Restart clock at recurrence; use risk of **latest resected** tumour.
- Persistent positive cytology with negative cystoscopy/imaging → biopsies, prostatic urethra, upper tracts.
- **UroVysion FISH** most specific marker; "anticipatory positive" precedes tumour. AUA: markers not instead of cytology; use for BCG response, atypical cytology.
- **Extravesical**: upper tract UC <1% low risk, ~10% high risk (13–18% after BCG), risk persists ≥15 yr. Prostatic urethra in high risk 20–40% at 10 yr; HG ductal → cystoprostatectomy ± urethrectomy.

10 · Prevention

- **Stop smoking** (recurrence effect uncertain); fluids, low-fat diet.

11 · Landmark trials

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline, suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine standard single instillation
SWOG maintenance (Lamm 2000)	Induction ± maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance essential
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

1 · Natural history and histology

- **20–30%** of new bladder cancers are muscle-invasive (MIBC); **~12%** node-positive or metastatic.
- **Primary MIBC** = invasive at diagnosis; **secondary** (**~20%**) = progressed from NMIBC → worse: more understaging, lower response to neoadjuvant chemotherapy (NAC).
- Most deaths after local therapy: within **2 years**, systemic (micrometastases) → add systemic therapy.
- UC = 90%. **Radical cystectomy (RC) is standard whatever the histology**; subtype changes only NAC/radiation use. Variants upstaged more often.

Variant	Key points	Treatment
Squamous (5%)	Schistosomiasis, chronic cystitis	Poor chemo/radiosensitivity → up-front RC + lymphadenectomy
Adenocarcinoma (2%)	Commonest in exstrophy; urachal; exclude colorectal/breast metastasis	RC (urachal: partial cystectomy)
Small cell	High stage; ectopic ACTH, hypercalcaemia	**Chemosensitive** → NAC etoposide + cisplatin, then RC
Micropapillary	Frequent LVI	Predominant → up-front RC
Mixed UC + variant	Quantify variant	UC-predominant → NAC; variant-predominant → RC

2 · Clinical staging

TUR → EUA → CT with excretory phase → chest. In cT2, pathology upstages **40–54%**, downstages ~17%; upstaging predicted by LVI, variant histology, female sex, secondary MIBC.

TURBT

- Resect all visible tumour if safe + separate deep specimen (aids NAC response, needed for preservation; pT0 at RC improves survival even without NAC).
- HG T1 or no muscle → repeat TUR within **6 weeks** (>40% upstaged without muscle).
- **Prostatic urethral biopsy** (trigone/bladder-neck tumour, CIS, multifocal): full-thickness loop mid-prostate to verumontanum at **5 and 7 o'clock** (densest ducts, commonest CIS). Stromal involvement affects NAC, urethrectomy, diversion choice.
- **Frozen urethral margin at cystoprostatectomy**: mandatory if TUR biopsy positive (neobladder only if negative); still do it if neobladder planned (higher NPV).

Bimanual EUA

Before and after resection, bladder empty. Both over- and (more often) understages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no 3-D mass
T3a	Mobile 3-D mass

cT	Bimanual finding
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before TUR** (post-resection inflammation mimics T3). CT nodal sensitivity only **30–50%**. Hydronephrosis → extravesical disease likely.
- MRI: poor renal function/contrast allergy; best local T-staging. **VI-RADS** (muscularis line): intact 1–3, interrupted 4, into fat 5; **≥3** suspicious (equivocal = positive).
- FDG PET/CT not routine (AUA: equivocal imaging, no biopsy possible); useless locally.
- Chest CT for all; bone/brain only if symptoms or raised ALP. Renal function decides NAC and orthotopic diversion.

3 · Pathological staging (AJCC 8th) and prognostic markers

Category	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac = **N3, not M1**; T4b N0 = stage IVA. pT4a needs **stromal** invasion (urethral/ductal CIS does not upstage). Organ-confined = ≤pT2bN0.
- **Subtypes**: luminal (FGFR3 mutations) best survival; **basal gains most from NAC**; p53-like chemoresistant. Not yet guiding treatment.
- **ctDNA**: >⅓ positive after RC, do badly; IMvigor010 adjuvant atezolizumab helped only ctDNA-positive.

4 · Radical cystectomy: principles and extent

- **RC + bilateral pelvic lymph node dissection (PLND) = standard for resectable MIBC**, ideally after cisplatin NAC. Delay > **12 weeks** → more extravesical/nodal disease, worse survival.
- **Male**: bladder, perivesical tissue, prostate, seminal vesicles. **Female** (anterior exenteration): bladder, urethra, uterus, cervix, ovaries, tubes, anterior vaginal wall. Steps → *Ch 132*; diversion → *Ch 133–135*.
- **Male organ-sparing**: no bladder-neck disease, negative prostatic biopsies; incidental prostate cancer ¼–½ → selected men.
- **Female organ-sparing** (no base/trigone tumour): uterus, anterior vagina → neobladder support, less prolapse; ovaries premenopause unless breast/ovarian risk; urethra unless bladder neck involved.

KEY POINT
Spare an organ only where oncologically silent: men – prostatic urethra and bladder neck; women – bladder neck, base, anterior vaginal wall.

5 · Pelvic lymph node dissection

- P4 Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — *templates; German trial and S1011 negative → standard PLND; extranodal extension*
- ~**25%** node-positive at RC; nodal status strongest survival predictor. Recurrence **70–80%** N+ vs ~30% N0. Rises with T (≤pT1 1–7% … pT4 up to 80%).
- Obturator and internal iliac = sentinel stations; skip metastases rare, common iliac involved 7–13%.
- Borders: medial ureter, lateral genitofemoral nerve, caudal Cloquet; cranial – standard to common iliac bifurcation/ureteric crossing, extended to aortic bifurcation (IMA controversial).
- ~25 nodes → ~75% detection; therapeutic threshold unproven; 10 nodes = usual pelvic-recurrence cut-off.

6 · Intraoperative decision making

- **cN+** → cisplatin chemotherapy first; responders → **consolidative RC** (14–25% pCR, much better survival).
- **Unexpected nodes**: frozen section; proceed only if completely resectable, within true pelvis.
- **Abort RC**: unresectable nodes, extensive periureteral disease, bladder fixed to sidewall, rectosigmoid invaded (most die within 2 years).
- **Ureteric margin**: involved 6–8%; positive → upper-tract recurrence (UTR 4–10%, mostly 2–4 years). Frozen section not mandated; send for gross tumour; CIS (skip lesions): resect maximally, keep diversion length. "Converted" margins still higher UTR; also prior BCG, CIS, hydronephrosis.
- **Male urethra**: recurrence 1–8%, mostly within 2 years; highest with **prostatic stromal invasion** (~30%) → consider urethrectomy (also diffuse prostatic urethral CIS). Positive pre-op prostatic biopsy does **not** bar neobladder: **frozen urethral margin decides** (specificity ~100%). LG papillary urethral tumour: resect pre-RC, keep urethra.
- **Female urethra**: involved 2–12% (bladder neck, base, anterior vaginal wall tumour). Urethrectomy mandatory only for T4 involving urethra/vagina. Proximal urethral frozen section in every neobladder candidate.

7 · Outcomes after RC

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** ultimately die of bladder cancer, mostly metastases within 2 years.
- **Pelvic recurrence**: low (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 + positive margins or <10 nodes) **39–46%**.

8 · Neoadjuvant therapy

- P2 Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types — *cisplatin NAC (ddMVAC 3–6/GC 4), rationale, pT0 ~90% RFS; adjuvant chemo (ABC 2022 +6%), nivolumab, Cairo radiotherapy*

Criterion	Cisplatin-ineligible (any one; ~50%)	Platinum-ineligible
Renal	GFR **<60 mL/min**	CrCl <30 mL/min
ECOG	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / ≥2	— / grade ≥3
Heart failure	NYHA III–IV	NYHA III–IV; multiple high-grade comorbidities

- NCCN prefers ddMVAC. Cisplatin-ineligible → straight to local therapy.
- **Neoadjuvant IO** (PD-1/PD-L1 blockade restores T-cell killing; anti-PD-L1: atezolizumab, durvalumab, avelumab; anti-PD-1: pembrolizumab, nivolumab). Phase II: **ABACUS** pCR 31%; **PURE-01** pT0 42%. Source: not yet recommended (NIAGARA, KEYNOTE-866, ENERGIZE pending). †NIAGARA since: perioperative durvalumab + GC improved EFS and OS.
- Neoadjuvant radiation: downstages, no survival gain.

9 · Adjuvant therapy

- Target: **pT3–4 or N+**. Post-RC morbidity blocks delivery in ~¼; neoadjuvant preferred (compliance, level 1).
- **Chemotherapy**: not standard; discuss ddMVAC/GC in **cisplatin-naive** pT3–4/N+.
- **Nivolumab 12 months**: ≥pT3/N+ after NAC, or cisplatin-ineligible/declining. IMvigor010 negative (included pT2).
- **Radiation**: pT3–4 local failure 20–40%; median survival after pelvic failure ~9 months. May consider for pT3/N+.

10 · Bladder preservation

Trimodality therapy (TMT)

- P4 Discuss bladder preservation surgery for TCC of urinary bladder — *selection (cT2, no hydronephrosis/CIS; 6–19%), sensitisers, schedules, outcomes, salvage RC 10–30%, follow-up; partial cystectomy*
- Concurrent chemotherapy beats radiation alone (two RCTs). Unfit for RC: looser criteria; extensive CIS and poor bladder relative contraindications.
- Needs MDT, compliant patient, salvage-RC counselling; balanced TMT vs RC discussion for fit patients.
- **Dose**: continuous to **60–66 Gy**, restage at 3–4 months (commonest); split course historical; **55 Gy in 20 fractions** non-inferior.

> **KEY POINT**
> TMT contains cystectomy rather than replacing it: salvage RC is the built-in safety net, so selection and lifelong cystoscopy decide success. Complete TUR cuts need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months); RETAIN: half recurred despite DNA-repair mutations
Radiation alone (64–66 Gy/32 or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging, ~25% occult nodes
Partial cystectomy + PLND	<3 cm solitary, no CIS, margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Late recurrence (lifelong follow-up). Frozen sections; RC if margin positive; consider perioperative cisplatin

11 · Metastatic urothelial carcinoma

- Bajorin adverse factors: KPS <80%, visceral metastasis (median survival 33, 13, 9 months with 0, 1, 2).
- **Cisplatin-eligible**: GC or ddMVAC; most respond, most progress. **GC = MVAC**, less toxic (von der Maase); dose-dense beat standard MVAC (EORTC 30924).
- **Cisplatin-ineligible, platinum-eligible**: gemcitabine–carboplatin (EORTC 30986) or enfortumab vedotin (EV) + pembrolizumab (NCCN preferred). †EV-302 later: EV + pembrolizumab ~doubled OS vs platinum in all comers → preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052) or atezolizumab if PD-L1+ (IMvigor210). Trap: **first-line IO alone only if platinum-ineligible.**
- **Maintenance avelumab** if no progression after platinum (JAVELIN Bladder 100). IO + chemo (IMvigor130, KEYNOTE-361, DANUBE) not recommended.
- **Second line**: **pembrolizumab** (KEYNOTE-045, regardless of PD-L1); alternatives nivolumab, avelumab; atezolizumab failed (IMvigor211). Platinum rechallenge if last dose ≥12 months earlier. Vinflunine Europe only.
- **ADCs**: **EV** (Nectin-4, MMAE; neuropathy, rash) beat chemo after platinum + IO (EV-301). **Sacituzumab govitecan** (Trop-2, SN-38; neutropenia) third line.
- **Erdafitinib** (oral FGFR inhibitor): FGFR3/FGFR2 alterations after platinum; response 40%; **hyperphosphataemia**, stomatitis.
- Failed: bevacizumab, sunitinib, trastuzumab.
- **IO toxicity**: rash, colitis, hepatitis, pneumonitis, endocrinopathies (often **irreversible**), myocarditis.

> **ALGORITHM**
> Platinum-eligible → GC/ddMVAC (or gemcitabine–carboplatin; or EV + pembrolizumab) → response/stable → avelumab maintenance → progression → pembrolizumab, or erdafitinib if FGFR-altered → EV → sacituzumab govitecan. Platinum-ineligible → pembrolizumab → EV.

12 · Follow-up

- **After RC**: CT with excretory phase + chest CT 6-monthly for 2–3 years, then yearly to 5 (EAU, AUA; NCCN 3–6-monthly for 2 years). Renal function each visit; B12 yearly from year 2;

cytology 6–12-monthly.

- Urethral wash (controversial) if CIS/prostatic urethral tumour or bloody discharge.
- Upper tract (4–10%) = commonest **late** recurrence; pelvic/distant mostly within 2 years (pelvic → chemoradiation, poor).
- Diversion checks: acidosis, renal function, stoma, hernia, stones → *Ch 133–135*.
- **After bladder preservation**: cystoscopy 3–4-monthly years 1–2, 6-monthly years 3–4, yearly to **10 years**; CTU + chest CT 3–6-monthly 2 years, yearly to 5.

13 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3 → RC vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 = surrogate; OS P = .06
BA06 30894 (EORTC/MRC)	CMV ×3 → RC/radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS HR 0.86; +5% at 5 years	Combination platinum only
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year PFS 66% vs 56%	ddMVAC better, more toxic
SWOG S1011	Standard vs extended PLND	No difference in DFS or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	DFS 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation ± mitomycin C/5-FU	Locoregional DFS 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (NS)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine–carboplatin vs MCAVI, cisplatin-ineligible	Equal OS, less toxicity	Gem–carbo standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard
EV-301	EV vs chemotherapy after platinum + IO	OS 12.9 vs 9.0 months	ADC third line

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

1 · Framing

- TUR stages everyone and cures low-grade NMIBC. MIBC → radical cystectomy (RC) + pelvic lymph node dissection (PLND), after cisplatin NAC if eligible → *Ch 131*.

2 · Transurethral resection of bladder tumour (TURBT)

Pre-operative evaluation

- **Continue aspirin**; hold warfarin, heparins, DOACs, P2Y12 inhibitors by half-life, bridging agreed beforehand.
- Bimanual; DRE if prostate suspected. **Treat active infection before resection**.
- Upper tracts: CTU, MRI, or ultrasound + retrograde. Iodinated contrast if GFR **≥60** mL/min; gadolinium if GFR **>30**.
- Orifice tumour → ureteroscope, stent, fluoroscopy ready.

Set-up and inspection

> **ALGORITHM**
> Lithotomy → antibiotic within 1 h (avoid fluoroquinolones) → 30° lens (urethra, orifices) → 70° (bladder neck, dome, anterior wall) → 120° if incomplete → map → blue-light/NBI if available (more tumours, less recurrence; CIS)

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Remove all visible tumour with **detrusor in specimen**; loop, cutting current; **~2 cm** margin of normal tissue where possible.
- Then EUA with bladder empty: **mass palpable after resection = cT3b**.
- † **En bloc resection**: one piece → orientation and detrusor kept, better T1 staging.
- Maximal resection: therapeutic in NMIBC; in MIBC going to RC no survival gain (residual tumour only prognostic) – clinical staging unreliable.

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk** → adduction → perforation	Low distension; bipolar; muscle relaxant (GA)
Ureteric orifice	Coagulation → stricture, hydronephrosis	**Pure cutting current**; routine stent not needed; short-term stent after large resection/heavy cautery reasonable
Dome	Intraperitoneal perforation; heat injury to bowel	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site / frequency	Away from dome; commoner	Dome, posterior wall; rare, large ones morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at same sitting	
Management	**Foley drainage + observation**	**Laparotomy**, inspect bowel, repair; catheter + drain

KEY POINT

Perforation can follow adequate resection of advanced tumour; it **rules out immediate intravesical chemotherapy** (extravesical leak).

Post-operative intravesical agents

- P3 Immediate post TURBT intravesical Therapy for TCC — *rationale, agents, contraindications*
- Reduce **recurrence**, not **progression**; only if no perforation.
- Single mitomycin C/epirubicin **within 24 h**: ~**13%** absolute fall in recurrence (meta-analysis). Gemcitabine 2 g (S0337): 12% at 4 years, no grade 4–5 toxicity.
- **BCG never immediately**: wait **2–4 weeks**, no gross haematuria († systemic absorption).

3 · Radical cystectomy: preparation

- **Stoma marking for every patient** (sitting, standing); consent for conduit.
- **No routine mechanical bowel prep** or oral antibiotics (faster recovery, fewer wound complications). **Alvimopan** (peripheral μ-opioid antagonist) 30–90 min pre-op then twice daily → less ileus, shorter stay.
- IV antibiotic within 1 h covering skin and bowel flora, e.g. **cefoxitin**.
- **VTE prophylaxis**: compression + pharmacological from **before induction**; extend **4 weeks** (28 vs 8 days → fewer major VTE, no extra bleeding). Enoxaparin standard; apixaban equally effective, better compliance.
- Men supine, flexed ~15°; women low lithotomy. Povidone-iodine 10%, no chlorhexidine on genitalia.

4 · Open radical cystectomy: technique

ALGORITHM

Lower midline → Retzius → urachus divided; peritoneum incised lateral to medial umbilical ligaments → vas/round ligament divided → caecum, small-bowel root, left Toldt mobilised → **retrosigmoid window** (left ureter) → retractor (check vena cava) → ureters freed with adventitia, obliterated umbilical artery divided for length, ureters divided → PLND → vesical pedicles stapled/clipped/sealed

- **Distal ureteric frozen section** controversial: positive margin predicts upper-tract recurrence, but clearing it neither prevents recurrence nor proves survival benefit.

Male

ALGORITHM

Peritoneum over seminal vesicles incised → rectum separated, Denonvilliers incised → posterior pedicles → endopelvic fascia → dorsal venous complex ligated → urethra divided

- After radiation/fibrosis: **sharp dissection under vision**. **Rectal injury → primary repair** ± flap ± faecal diversion.
- **Urethral frozen section** if neobladder planned: very high NPV (prostatic urethral biopsy predicts poorly). **Positive → no orthotopic neobladder** (warn beforehand).
- **Nerve sparing**: few potent at 1 year (age predicts); local recurrence even in pT2; avoid if advanced or pre-existing ED.
- **Prostate sparing**: better potency/continence, but occult prostate cancer common, distant metastasis frequent; needs normal DRE/PSA, TURP and negative frozen section.

Female

- Standard = **anterior exenteration** (essential if positive EUA or cT4a). Women present later; female sex independently predicts cancer death.
- Counsel on sexual morbidity; ovaries make testosterone after menopause → discuss **ovarian sparing** if not locally advanced.

> **ALGORITHM**
> Ovarian vessels ligated → posterior fornix incised → pedicles → urethra excised from above or introitus → **clamshell vaginal closure** (keeps girth, loses length), watertight 2-0 polyglactin → pack out day 2

- Sinusoidal vaginal pedicles → meticulous haemostasis; leaky closure → peritoneal fluid per vaginam.
- **Vagina + urethra sparing for neobladder**: ≤cT2, uninvolved bladder neck, negative urethral frozen section; plane on pack/EEA sizer; divide pedicles only to mid-lateral vaginal wall; preserve striated sphincter.

5 · Pelvic lymphadenectomy

Template	Boundaries	Evidence
Standard	Lateral genitofemoral nerve; medial internal iliac artery; distal Cooper ligament; proximal ureter crossing common iliac	Reference
Extended	+ common iliac and presacral	Better staging; possible survival gain
Superextended	Up to IMA (preaortic)	No benefit over extended

- Protect obturator nerve; seal lymphatics. **≥10 nodes** → better survival (NMIBC at RC: >20).
- RCT: **extended not superior to limited** for RFS/CSS; staging value certain (finds extra-pelvic N+ for adjuvant therapy).

6 · Partial cystectomy

- **Solitary small MIBC without CIS**; **urachal adenocarcinoma** (en bloc umbilicus + urachus + dome, frozen margins). Same oncology: PLND, NAC if eligible, consent for RC. Muscle-invasive recurrence → salvage RC.

> **ALGORITHM**
> PLND → cystotomy away from tumour → full-thickness excision with fat, 1–2 cm frozen-section margin → reimplant ureter if needed → 2–3-layer watertight closure → warm water irrigation (seeding) → drain → cystogram day 7, remove catheter

7 · Robotic radical cystectomy (RARC)

KEY POINT
No effective salvage exists: urine spillage, poor PLND or positive soft-tissue margin → near-uniformly fatal recurrence, so robotics is judged on oncology first.

- Indications as open; Trendelenburg + pneumoperitoneum strain heart/lungs; prior surgery, obesity relative contraindications.
- Xi side-docks from right (older systems between legs). Ports higher than prostatectomy (extended PLND, intracorporeal diversion); **15-mm left assistant port** for stapling, bagging packets.

ALGORITHM
Sigmoid released → presacral plane → cul-de-sac/prostate apex → Retzius → anterior pedicle → ureters freed (**never grasp**) → **extended PLND with ureters intact** → ureters clipped, margin sent, left ureter under sigmoid → posterior pedicles → DVC → urethra (clip proximally only if neobladder)

- PLND before ureteric division keeps urine output as resuscitation guide. Bag each packet whole; obturator nerve is **not** the posterior limit – go to hypogastric vein. Female: EEA sizer for countertraction.

8 · Complications and post-operative care

- Complications common, few major; GI, then infection.
- **Ileus**: alvimopan, neostigmine (telemetry), early feeding, opioid-sparing analgesia; **no routine NG tube**. VTE prophylaxis, mobilisation, spirometry.
- Ureteric devascularisation → ischaemic stricture (keep adventitia); vaginal leak/fistula ← non-watertight closure. Diversion complications *→ Ch 133–135*.

9 · Landmark trials

Trial	Design	Result	Take-home
Grossman 2003	NAC + RC vs RC	Median survival 46 → 77 months	Offer cisplatin NAC
SWOG S0337	Single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low grade
RAZOR (2018)	Non-inferiority RARC vs open; open diversion	2-year PFS 72.3% vs 71.6%; less blood loss, shorter stay	Robotic oncologically equivalent
iROC	RARC + intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE, wound complications	Recovery advantage

133 Use of Intestinal Segments in Urinary Diversion

1 · Surgical anatomy

- **Stomach**: right gastroepiploic (gastroduodenal ← hepatic) and left gastroepiploic (splenic); greater-curve arcade carries the pedicle. Pedicle = antrum or fundal wedge; left gastroepiploic occasionally atretic → use right. Fundal wedge must **never reach the pylorus or lesser curve**; antrectomy → Billroth I. Thick seromuscular layer suits a submucosal ureteric tunnel.
- **Small bowel** ~22 ft: proximal 2/5 jejunum (wider, single arcades, thin mesentery), distal 3/5 ileum (multiple arcades). Do not rely on more than **8 cm** of bowel surviving away from a straight vessel → clear ≤8 cm of mesentery at a segment end.
- **Pelvic small bowel – avoid after radiotherapy**: last 2 inches of terminal ileum (fixed) and the 5 ft starting ~6 ft from Treitz.
- **Colon**: SMA (ileocolic, right colic, middle colic); IMA (left colic, sigmoid → superior haemorrhoidal); internal iliac → middle haemorrhoidal; linked by the **marginal artery of Drummond**. Right colic from ileocolic → ascending colon reaches pelvis easily.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic ↔ ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid ↔ superior haemorrhoidal

> **KEY POINT**
> Watershed zones are fed by the smallest terminal branches of two arteries → ischaemic first in hypotension/embolism → site anastomoses to one side.

2 · Segment selection

- P2 P1 Use of intestinal segments in urology / Intestinal segments in urology — *segment table (electrolytes, pros, cons, when to choose), NH_4Cl mechanism, alkali therapy*
- Neobladder reach: caecal mobilisation, mesenteric Z-plasty, bed flexion.
- **Stomach** (renal failure, short gut, dense adhesions): avoid antrum if fundus available; consider long-term H2 blocker/PPI. **Haematuria–dysuria syndrome**: spasm, pain, haematuria/dysuria without infection, skin excoriation; intermittent; H2 blockers. Bacteriuria no lower than ileum/colon. Gastric-surgery sequelae: dumping, small-stomach and afferent loop syndromes, anaemia.
- **Jejunum**: only if nothing else; use most distal. Colon: obstruction needing reoperation ~5% (half ileal rate).
- Bowel loss: **ileum** → B12, bile-salt and fat malabsorption (loss of ileal brake); **ileocaecal valve** → bacterial overgrowth, vitamin A/D deficiency, fast transit, bicarbonate/fluid loss; **colon** → diarrhoea, dehydration.
- **B12**: liver stores last **3–5 yr**; clinical deficiency rare; markers homocysteine, methylmalonic acid; empirical parenteral B12 after 5–7 yr of ileal diversion.
- Clam cystoplasty → more bowel dysfunction (overactivity associates with irritable bowel) → warn.

3 · Bowel preparation and enhanced recovery

- P3 Methods of preoperative bowel preparation — *mechanical lowers amount, antibiotics lower concentration; oral regimens and harms; IV prophylaxis most important; ERAS; who still needs full prep*
- Oral antibiotics cut concentration to ~10^2/g; descending colon **$10^{10\text{-}12}$/g**. Kanamycin = best single agent.
- Colorectal data caveat: urology opens long segments into a sterile tract.
- **IV antibiotic most important**: 1–2 h before incision, covering **anaerobes**. Authors: clear liquids, magnesium citrate, oral neomycin the day before.
- **ERAS**: **no routine bowel prep**; does not raise complications/readmissions; improves opioid use, diet, ICU time, QoL. Gum, metoclopramide help.
- Postoperative ileus (POI) = commonest complication of cystectomy (more than colectomy: deperitonealisation, urine exposure, long surgery).
- **Alvimopan**: peripheral μ-opioid antagonist; 12 mg 30 min–5 h pre-op, then twice daily ≤7 days (15 doses); inpatient only.
- Salvage or benign cystectomy (radiation, fistula, neurogenic): full mechanical + antibiotic prep, slow diet.

4 · Intestinal anastomoses

Bowel complications = commonest cause of early morbidity and mortality after diversion.

1. **Exposure**, wall off; **blood supply** (transilluminate, bleeding edges, no tension).
2. **Prevent spillage**: non-crushing clamps; irrigate segment.
3. **Serosa-to-serosa, inverted**, watertight; **no strangulating sutures**; **mesentery aligned**.

- Breakdown: ischaemia, faecal sepsis, **drain on the anastomosis**, irradiated bowel (most lethal bowel complications). Sutured = stapled; stapled side-to-side ileoileal commonest.
- Leak presents late (2nd week). Neobladder–vaginal fistula: continuous incontinence; vaginal-sparing cystectomy reduces it.
- **Obstruction** ~10% (ileum/stomach), ~5% colon; adhesions > recurrent cancer. Prevent: unirradiated bowel, close apertures, omentum, **reperitonealise**. Total exenteration → polyglactin mesh + omentum for pelvic floor.
- **Ogilvie syndrome**: dilated caecum; water-soluble enema excludes obstruction; caecum **>12 cm** or rising white count → neostigmine/colonoscopic decompression; surgery for ischaemia, peritonitis, refractory.

5 · Ureterointestinal anastomoses

- Minimal mobilisation; preserve periadventitia; spatulate; mucosa-to-mucosa; **bring bowel to ureter**; retroperitonealise; soft stents (leak 3–5% → near zero).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm)	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%

Technique	Type	Key feature	Stricture	Leak
Wallace	Refluxing end-to-end	Conjoined ureters (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

- Wallace: spatulate 1.5–2 cm; easy endoscopic access; tumour blocking both ureters exceedingly rare. Best antireflux: colon – Pagano; small bowel – Le Duc, serosal apposition.

> **KEY POINT**
> Conduit pressure is damped by ureteric peristalsis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop colonisation → stented refluxing anastomosis is default; reflux allows loopogram imaging of upper tracts.

- Urinary fistula days 7–10 (3–9%) → fibrosis → later stricture.
- **Stricture**: ischaemia, leak, radiation, infection; lifelong surveillance. Classic: **left ureter under the IMA**.
- Open reanastomosis most durable; endoincision ~50–60% at 2 yr; balloon not durable. Poor endourology: stricture <1 yr post-op, **≥1.5 cm**, left-sided. Metal stents only for limited life expectancy.

6 · Conduit diversion

- **Ileal conduit** – simplest, fewest early complications. Avoid in short bowel, inflammatory small-bowel disease, irradiated ileum.

> **ALGORITHM**
> Segment **10–15 cm**, 10–15 cm from ileocaecal valve → keep major arcade, windows ~5 cm → ileoileostomy, segment caudal → close mesenteric window → flush → left ureter behind sigmoid mesentery → stented anastomoses → fix base retroperitoneally, RLQ → stoma

- Missed duplex ureter → intraperitoneal leak. Stomal bleeding ~10%; intractable → varices of portal hypertension → TIPS/embolisation.
- **Jejunal conduit**: 15–25 cm from Treitz, LUQ stoma; only when ileum and colon unusable; contraindicated in nutritional bowel disorders.

	Transverse	Sigmoid	Ileocaecal
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy	Long ureteric replacement
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- Colon conduits: avoid in IBD, chronic diarrhoea.
- **Ileal vesicostomy**: neurogenic, best with DSD; reversible.
- Stents out days 4–6; drain retroperitoneally, **never intraperitoneally**.

Complication (%)	Ileal conduit	Colon conduit	Continent
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27

Complication (%)	Ileal conduit	Colon conduit	Continent
Renal deterioration	18	15	—

Gastric conduit: acidosis 0%.

7 · Stomas

- Commonest problem after diversion; mostly technical.
- Siting: mark sitting and supine; **through rectus** (lateral → hernia); ≥5 cm from incision; avoid creases, scars, umbilicus, belt line, irradiated skin.
- Aperture: **keep subcutaneous fat**, cruciate fascial incision, admits two fingers.
- **Nipple (rosebud)** for appliance (less stenosis, better fit): 5–6 cm out for a 2–3 cm spout; eversion sutures to subdermis, never full-thickness skin. **Flush** for continent catheterisable (low profile).

	End stoma	Loop (Turnbull)
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%
Best for	Standard	**Obese** (thick, short mesentery)

- Hernia (ileal conduit 2–6.6%); stenosis (ileal 20–24%, colon 10–20%; paediatric catheterisable >50%). Pseudoverrucous lesions → acidify urine, fit appliance.

8 · Renal function

- Deterioration after conduits 10–74% (ileal conduit 18%); ~6% die of renal failure. Ureterosigmoidostomy: renal failure leading cause of death after 15 yr.
- Continent > conduit demand (length, dwell time). **Tubular acidification matters as much as GFR.**

Test	Threshold for continent diversion
Creatinine + urine protein	**<2.0 mg/dL**, no proteinuria
GFR	**>35 mL/min**
Ammonium chloride load	Urine pH **≤5.8**
Water deprivation	Osmolality **≥600 mOsm/kg**

- Exception: retentive diversion in ESRF children awaiting transplant. **Bowel invalidates** urine osmolality, pH, serum urea and creatinine.

9 · Metabolic complications

Absorption depends on **segment, surface area, dwell time, solute concentration, renal function, pH**.

Segment	Na	K	Cl	pH	Associated	Treatment
Stomach	—	↓	↓	↑	↑ Aldosterone; lethargy, seizures, arrhythmia	H2 blocker, PPI; arginine HCl; remove segment
Jejunum	↓	↑	↓	↓	↑ Renin, angiotensin; vomiting, dehydration	IV saline, bicarbonate, thiazide

Segment	Na	K	Cl	pH	Associated	Treatment
Ileum/colon	—	↓	↑	↓	Total-body K depletion; hypocalcaemia	Alkali; chlorpromazine, nicotinic acid

KEY POINT
Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na, Cl; high K); ileum/colon absorb NH_4Cl (high Cl). **Only jejunum is hyperkalaemic**.

- **Gastric alkalosis**: usually trivial; severe with renal failure/dehydration (vomiting → aldosterone ↑ → bicarbonate retained). **Gastrin** above **120 ng/L**: small rises → large bicarbonate rises; distension stimulates gastrin. Rehydrate + H2 blocker → arginine HCl if life-threatening → omeprazole → remove segment.
- **Jejunal syndrome**: NaCl loss → hypovolaemia → renin–aldosterone → low-Na, high-K urine → cycle. Worse with proximal segments, parenteral nutrition; can be fatal. Long term oral salt, thiazide.
- **Hyperchloraemic acidosis**: worst with **ureterosigmoidostomy** and colonic pouches (most Mainz patients need alkali); NH_4^+ also enters via K^+ channels. Villous atrophy but transporters persist.
- Alkali: sodium bicarbonate (gas); Bicitra/Shohl's (poor taste); Polycitra when K needed and Na unwanted. **Chlorpromazine 25–50 mg tds** (tardive dyskinesia) or **nicotinic acid 400 mg 3–4×/day** (avoid in ulcer, liver disease) inhibit cAMP → limit, not correct.
- **Hypokalaemia**: commoner with colon (ileum reabsorbs K). **Trap**: correcting acidosis alone → profound hypokalaemia, flaccid paralysis → give K with bicarbonate. Also low Mg, Ca.
- **Altered sensorium**: **ammonia** (commonest; liver dysfunction + ureterosigmoidostomy), Mg deficiency, drugs. Treat: drain diversion → oral neomycin → protein restriction → IV arginine glutamate 50 g → lactulose.
- **Drugs reabsorbed**: **phenytoin, some antibiotics, methotrexate → drain continent pouches during chemotherapy**.
- **Osteomalacia** (commonest with ureterosigmoidostomy): acidosis (bone buffers H^+), vitamin D resistance, renal Ca loss (sulphate), PTH resistance. ALP ↑. Correcting base deficit >2.5 mEq/L prevents bone loss; alkali + calcium → active vitamin D.
- **Infection**: ~3/4 conduit urines bacteriuric; **treat pure Proteus or Pseudomonas**; observe asymptomatic mixed cultures.
- **Stones** (struvite/Ca phosphate; acidosis, urease organisms, **staples**, mucus): colon conduit 3–4% < ileal conduit 10–12% < continent caecal reservoir ~20%.

Cancer

- P4 Secondary tumours after urinary diversion — *Kälble risk table by diversion, surveillance, pathogenesis, adenocarcinoma at ureterocolic junction*
- **Ureterosigmoidostomy**: 6–29% (mean 11%) after **10–20 yr**; defunctioned anastomoses still develop cancer → **excise, do not ligate**.

10 · Neuromechanical aspects

- † Detubularisation into a sphere disrupts coordinated peristalsis and, by Laplace's law, increases volume and lowers pressure – basis of reservoirs and neobladders.

11 · Landmark trials

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

134 Continent Urinary Diversion

1 · Definition and basic principles

- P4 P2 Continent urinary diversion: orthotopic vs non-orthotopic / Principles of continence in continent diversion — *definition, three principles (reliable outlet, low pressure by detubularisation + sphere/Laplace, capacity ≥500 mL; 44 cm ileum <200 → 500–600 mL in a year), ileum stretches more than colon*
- Orthotopic now preferred (natural voiding, no appliance, fewer revisions); cutaneous for pre-existing incontinence or urethral disease. Detubularisation cuts the circular muscle; Kock double-folding, S/W shapes.

KEY POINT

Continence is a pressure balance: low reservoir pressure (detubularisation) against adequate outlet resistance (sphincter or efferent valve). Failure of either → leakage. Camey I (tubular ileum) failed because its pressure was too high.

2 · Patient selection

- P4 Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion — *eligibility 80–90% men / ~75% women, counsel all options (AUA/EAU), absolute and relative contraindications table, renal thresholds, ~10% intraoperative change, age/obesity not bars*

ALGORITHM

eGFR >40 → intact external sphincter → adequate bowel → not frail/no severe liver, cardiac or psychological disease → men: no cancer at distal prostatic urethra; women: no palpable anterior vaginal tumour or cervical invasion → willing to self-catheterise → offer orthotopic

- Mark a conduit site in everyone (free of creases standing/sitting/lying; away from umbilicus and scars; as lateral as possible while through mid-rectus). Authors prefer umbilicus for a continent cutaneous stoma.

Urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually N+ → neoadjuvant chemo)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op	TUR biopsy 5 and 7 o'clock beside verumontanum — optional	EUA; bladder-neck biopsy
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; AUA requires negative margin)	
Bladder CIS	Not predictive	
Skip lesions	—	None: negative bladder neck → negative urethra

- Risk: none < CIS/mucosal/ductal < stromal. **Neobladder is protective** (HR 0.44). TUR misses much prostatic disease → not repeated just for this; authors send a urethral margin from the final specimen.
- Prostatic involvement → urethral surveillance (wash cytology, urethroscopy). Persistent prostatic urethral disease → cutaneous diversion + urethrectomy.

- Women: vaginal invasion always with bladder-neck disease; preserve urethra if bladder-neck biopsy and proximal urethral frozen section negative. Preserve the female urethra whatever the diversion (distal urethrectomy devascularises the clitoris).
- **Locally advanced/N+ at surgery is not a contraindication**: pelvic recurrence ~7% (13% if extravesical/N+), rarely disables the neobladder; exception grossly positive margin at pubis, rectum or side wall. Adjuvant RT 45–50.4 Gy to a neobladder well tolerated.

Patient-related factors

- **Renal function** — key contraindication: creatinine **< 1.7–2.2 mg/dL** or eGFR **> 35–40 mL/min**. eGFR 40–59 is safe (decline no faster than conduit); obstructive uraemia often reverses. All: hydration, avoid nephrotoxins, oral bicarbonate. *→ Ch 133*
- Commonest technical reason to abandon orthotopic = short/fatty mesentery. Reach: deeper distal mesenteric division, discard proximal segment, less table flexion/Trendelenburg, gentle traction.
- **Self-catheterisation**: 10–20% men, ~30% women with neobladder; all cutaneous pouches; may appear years later → all must be willing. MS, quadriplegia, frail → conduit.
- **Sphincter weakness**: man → artificial urinary sphincter; woman → Burch/sling (likely catheterisation).
- **Prior radiation**: feasible if no incontinence, stricture, fistula or severe bowel damage; brachytherapy/high-dose RT and cervical implants damage the rhabdosphincter most; final decision intraoperative.
- **Prior prostatectomy**: pre-cystectomy continence predicts outcome; flexible cystoscopy locates old anastomosis. **Prior bowel resection**: reuse the old anastomosis as one reservoir end; short bowel → sigmoid neobladder.

3 · Preserving continence at cystectomy

Rhabdosphincter (pudendal nerve) is critical; lies **anterior and lateral** to the urethra in both sexes.

Men

- Posterior dissection never distal to apex. Minimal endopelvic fascia and puboprostatic incision; DVC with **absorbable** suture (clips/staples erode into anastomosis).

> **ALGORITHM**
> Endopelvic fascia → slightly incise puboprostatic ligaments → ligate DVC close to apex → incise anterior urethra → place sutures → posterior sutures through rectourethralis → divide posterior urethra → tag until after ureteroenteric anastomosis

- Sutures placed **before** full division (male urethra retracts). Apical frozen section: tumour → urethrectomy + cutaneous/conduit; atypia/focal CIS → second distal margin.
- Nerve sparing: continence benefit controversial; preserves membranous urethral sensation → reflex sphincter closure (night continence).

Women

- Spare uterus, anterior vagina in selected low-stage disease → no fistula, better sexual function, less retention. Keep vaginal dissection no more dorsal than 2 and 10 o'clock. Palpable bladder-neck/vaginal tumour → en bloc urethrectomy + cutaneous diversion.

ALGORITHM
Posterior pedicles → plane off anterior vagina → Foley balloon marks junction → free bladder neck and first 0.5 cm urethra → clamp bladder neck → divide just distal → frozen section

- Endopelvic fascia untouched; bladder neck not needed for female continence; urethra does not retract (sutures after removal).
- Always interpose an **omental flap** between vagina and pouch. Authors add sacrocolpopexy. If vaginal wall excised, keep vaginal and urethral suture lines apart.

4 · Choice of bowel and ureteric reflux

- Distensibility: ileal circular > colonic circular > ileal longitudinal > colonic longitudinal. **Ileum default**; mucosa atrophies (less mucus, absorption). Colon has pressure waves. Stomach: poor compliance, haematuria–dysuria.
- Avoid deep proximal mesenteric incision (reservoir supply). Absorbable closure — staples cause stones.
- **Antireflux unnecessary**: Valsalva raises pressure equally; every valve can obstruct. Direct (Wallace/Leadbetter) stricture **3–6%** vs Le Duc 7–29%. **Obstruction, not reflux, destroys kidneys.** Kock nipple failures are late (>10 yr), silent until bilateral hydronephrosis.

5 · Orthotopic neobladder designs

- P2 P4 Use of intestinal segments in urology, and orthotopic neobladder — *design table (Studer, Hautmann, T pouch, Abol-Enein, Camey II, Hemi-Kock, Padua, Mainz III, sigmoid); Studer steps; Hautmann vs Studer*
- Studer most popular. Hautmann refluxing since 1997 (stenosis 9.5% → 1%); a 40 cm version limits retention and reabsorption; 5 cm U-flap at urethral point. Hemi-Kock abandoned. Padua (VIP) used robotically. Sigmoid: worst night continence.
- **T pouch**: distal 3–4 cm afferent limb in serosal trough fixed through mesenteric windows, covered by ileal flaps; tapering abandoned (ostial stenosis).

ALGORITHM
Studer: ileum 15–20 cm proximal to ileocaecal valve → discard ~5 cm proximally → 40–44 cm reservoir (two 20–22 cm limbs) + afferent limb → open, oversew back wall → fold opposite way → stented end-to-side ureteroileal → urethral anastomosis at dependent point

KEY POINT
Hautmann vs Studer = capacity trade-off: bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · Continent cutaneous reservoirs

- P2 Principles of continence in continent urinary diversion; Mitrofanoff principle — *nipple vs flap vs imbricated valve, appendix ≥5–6 cm/≥12 Fr, Mitchell extension, Yang–Monti, tapered ileum, Indiana*
- Nipple valves (Kock) historical. **Umbilical stoma**: less stenosis, easiest for paraplegics, near-invisible (medical alert bracelet). Lower quadrant: concealed. Appendix takes only 14–16 Fr catheters.
- **Indiana**: right colon + ~10 cm terminal ileum; valve double-imbricated with non-absorbable Lembert sutures over 3–4 cm on a 16 Fr catheter; ileum tapered. Whole colon opened because residual caecal peristalsis overcame the valve.

- **Right colon pouch** (authors): ~30 cm colon; intact terminal ileum as afferent limb (ileocaecal valve antireflux); appendix in 4–5 cm taenial trough → umbilicus. Unsuitable appendix → Yang–Monti.
- Intra-op: fill, check continence and catheterisation; fix reservoir to abdominal wall.

7 · Perioperative care

- **ERAS**: prehabilitation, no oral mechanical prep (authors still prepare for colon), carbohydrate loading, restrictive fluids, no NG tube, early diet, opioid-sparing; **alvimopan** cut ileus ~23% → ~7%. Ileus = commonest GI complication. Up to 70% malnourished/frail.
- **Drainage**: 7 Fr single-J stents + 24 Fr urethral catheter, irrigated for mucus; catheter out at **3 weeks**; pouchogram only if drain output significant. Then timed voids, lengthened progressively, + pelvic floor exercises. Robotic → *Ch 135*.

8 · Results and complications

Early (<90 days) morbidity not higher than ileal conduit; colonic cutaneous has more infection and dehiscence.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; undrained → percutaneous drain ± nephrostomies; **avoid early open repair**
Bacteriuria	Very common	Treat only if symptomatic or *Proteus/Pseudomonas*; febrile UTI → look for obstruction, stones, residual; recurrent → nitrofurantoin/methenamine
Ureteroileal stricture	Can appear after 10 yr, often silent	CT/US → pouchogram/antegrade; endoscopy (retreatment >50%) inferior to open
Pouch stones	Kock and stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention	Cystogram; catheter ± drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; beyond first weeks won't heal → transvaginal/transabdominal repair or cutaneous conversion
Stomal stenosis (cutaneous)	Up to ~20%; after 90 days, radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Endoscopy for level; below fascia → explore
Incisional hernia	Common; impairs emptying	Repair

Continence

- Day continence over **3–6 months**, ultimately **80–90%**; night lags into year 2, persistent nocturnal incontinence **20–50%**. Risk: age >65, colon, diabetes, no nerve sparing, hysterectomy.
- Night leak: no sphincter–detrusor reflex, lower nocturnal tone, bowel secretes water → void at bedtime, alarm 2–3 times.

- Evaluate after 12–18-month plateau; exclude fistula; biofeedback; urodynamics for pressure waves (colon).

Stress incontinence	Men	Women
Bulking agent	Marginal	Mild only; <50%, poorly durable
Definitive	AUS or male sling	Pubovaginal sling — most then catheterise
Safety rule	No blind retropubic needle passage → bone anchors or prepubic route	

- Persistent cutaneous stomal leak → surgical revision of efferent mechanism; minor → bulking.

Retention

- Presents as infection, overflow, palpable pouch, new hydronephrosis. Men 4–10% (~21% by 10 yr; long bowel, prostate-sparing). **Women 20–30%** (up to ~60% self-catheterise on questionnaires).
- Female cause **mechanical**: pouch falls back, angulating pouch–urethral junction. Prevent: omentum behind pouch, vaginal suspension, sacrocolpopexy, uterine preservation.
- Treat with CISC (drugs useless); exclude stricture (~1%) or recurrence; obstructing mucosal flap → endoscopic incision.

> **KEY POINT**
> Men fail by **leaking** (sphincter without reflexes) → augment sphincter. Women fail by **not emptying** (unsupported pouch kinks) → support the pouch.

9 · Follow-up

- Phases: first 4 months → stricture; 4 months–3 yr → recurrence; >3 yr → stones, late obstruction, urethral/upper-tract tumour.
- 4-monthly year 1, 6-monthly to 3 yr, then annually: examination, metabolic panel, blood count; CT + chest imaging to 5 yr.
- Annual B12; urethral wash cytology in men (esp. prostatic CIS); bimanual + voided cytology in women (urethral recurrence invisible on CT). After 5 yr lifelong upper-tract US, metabolic panel, blood count.
- QoL: no diversion proven superior; neobladder better body image, conduit better urinary function.

10 · Landmark trials

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 RC patients	No difference in renal function or infection at 3 yr; T pouch more secondary procedures	Antireflux adds cost without benefit
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

135 Minimally Invasive Urinary Diversion

1 · Context

- Robot-assisted radical cystectomy (RARC) is oncologically equivalent to open in RCTs (RAZOR) — but with **extracorporeal** diversion (ECUD); intracorporeal (ICUD) untested.
- IRCC: ICUD rose ≈**11%/year**, driven by **ileal conduit** (commonest robotic diversion).
- Decline of continent reservoirs predates robotics; no good evidence they give better QoL.

2 · Patient selection

High BMI, age, locally advanced disease and prior surgery no longer exclude. Three separate decisions:

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary tolerance of steep Trendelenburg (relative; eased by Xi and AirSeal)
Use of ileum	Short bowel; severe IBD; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function — as open

ALGORITHM

Pasadena Consensus learning pathway: RARC + extracorporeal diversion → intracorporeal conduit (≈30–100 cases) → continent (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · Perioperative bundle

ERAS: alvimopan, goal-directed fluids, immediate oral diet, opioid-sparing analgesia (paracetamol/ketorolac, not epidural/PCA) → less ileus, shorter stay. Caution: stay **<5 days** ≈ doubled 90-day major complications.

Element	Recommendation
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, no oral antibiotics (electrolyte upset; liquid spillage; no benefit)
Thromboprophylaxis	Pneumatic compression + stockings; LMWH before and up to **4 weeks** after
Antibiotics	Broad-spectrum IV within **1 h**

4 · Position and ports

- Lithotomy, steep Trendelenburg (Xi table allows flatter); Hasson entry if prior surgery; 0° lens (30° down for deep female pelvis).
- **6–7 ports placed more cephalad** (reach over small bowel, nodes to aortic bifurcation): camera 12 mm above umbilicus; 8-mm arms lateral to rectus; fourth arm above/medial to ASIS; extra 12–15-mm left port for stapling (neobladder); assistant ports; suprapubic 12 mm for bowel reanastomosis, extended to Pfannenstiel for specimen.
- Undock and reduce Trendelenburg for the urethra–neobladder anastomosis.

5 · Nerve sparing

Same autonomic plexus serves sexual function and neobladder continence → nerve sparing is a continence operation. Select safe tumour, low burden, good baseline function.

	Male	Female
Danger points	Presacral/medial common iliac → high pedicle ligation → seminal vesicles/Denonvilliers → prostatic pedicles, apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus)
Manoeuvre	Athermal, intrafascial, traction-free pedicle control	Preserve tissue around proximal urethra, anterior vaginal wall; extract via Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by DVC; dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third; continence survives loss of bladder neck

6 · Intracorporeal ileal conduit

ALGORITHM

Left ureter under sigmoid mesentery to right → 12 cm ileum, 15–20 cm from ileocaecal valve → marionette stitch → isolate with 45-mm stapler → ureteroileal anastomoses (left first) + stents → bowel reanastomosis → stoma.

- **Marionette stitch**: 1-0 silk through hypogastrium and stoma end, held **untied** — replaces the missing hand, positions the conduit.
- Avoid redundant segment → *Ch 133*. Tag cut ends (prevents malrotation); washout abandoned.
- **Enterotomy**: athermal cold-scissors buttonhole with serosal removal (thermal slit → more strictures). Retroperitonealise anastomosis.
- Stents: 8.5-Fr single-J; **guidewire removed only after anchoring**; ends out via 15-mm port.
- Bowel restored after both ureters. Bricker (separate end-to-side) or Wallace (Y, one anastomosis).
- Stoma: clamp through site grasps marionette and stents → undock → deliver conduit; check orientation.

7 · Intracorporeal neobladder

Key difference: **early neobladder–urethral anastomosis** (Karolinska) — anchor dependent ileum to urethra first, then build pouch in situ. Design principles → *Ch 134*.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded
UCLH Pyramid	50 cm	Koie-derived cross-fold
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	One trough at a time; chimneys

ALGORITHM
Urethral anastomosis (Van Velthoven barbed suture) → detubularise distal 40 cm (cold scissors) → posterior plate → ureters to afferent limb/chimney, stented → close anterior wall after 22-Fr catheter confirmed.

- Bowel restored via suprapubic port: 60-mm side-to-side stapled; close mesenteric window.
- **Tension-free urethral anastomosis** (no tactile feedback): maximal urethral length, sigmoid out of pelvis, less Trendelenburg, perineal pressure, barbed suture, Penrose traction (Karolinska), Rocco-type stitch (USC). Incising mesenteric peritoneum risks pouch ischaemia.

Intracorporeal Indiana pouch

ALGORITHM
Mobilise right colon → divide ileum 15 cm from ileocaecal junction and transverse colon proximal to middle colic → restore ileocolic continuity immediately → detubularise along anterior taenia, fold → ureters through posterior plate, double-J; 24-Fr caecostomy via appendiceal stump → taper ileum over 14 Fr, imbricate valve → umbilical stoma.

KEY POINT
Capacity from detubularised right colon; continence from tapered ileum + imbricated ileocaecal valve.

8 · Postoperative care

Avoid overhydration; no routine NG tube (Cochrane). Irrigate neobladder **8-hourly** (mucus).

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Neobladder catheter	**3–4 weeks**, after leak-free pouchogram
Externalised stents / double-J	≈1 week / cystoscopically 3–4 weeks
Feeding-tube stents tied to Foley	With catheter

9 · Outcomes

- **Operative time** = quality measure; strongest determinant diversion type (conduit ≈70 min shorter); ICUD ≈ ECUD time. Proficiency ≥ **66 RARCs**.
- **Complications**: mostly diversion-related, infection commonest; two-thirds of reoperations after 3 months → 90-day reporting underestimates. RCT meta-analysis: no difference in 90-day major complications.
- **Parastomal hernia** ≈one-third by 3 yr (long operation, fascial defect ≥30 mm, low eGFR); robotic mesh repair durable.
- ICUD vs ECUD (IRCC): adjusted, neutral for high-grade complications/readmission; less transfusion.
- **Reoperation** ≈half by 5 yr; predictors: neoadjuvant chemotherapy, any high-grade complication.
- **Uretero-enteric stricture** rises over 5 yr; robotic revision = open; ICG localises ureter.

KEY POINT

Night continence is worse because the urethrosphincteric guarding reflex is inactive in deep sleep → continence depends on reservoir capacity (detubularisation, nerve/sphincter preservation, night alarm, limit nocturnal output). Assess at 6–12 months.

- Good continence: good ECOG status, retained filling sensation. Up to 25% of men need CIC after open neobladder.
- Intracorporeal: day continence 68–100%, night 55–85%, CIC up to 11%. Emptying and night continence lag; only FloRIN matched open on urodynamics.
- QoL: physical domains recover by 24 months except sexual function.

10 · Landmark trials

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD
iROC	Multicentre RCT, intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early, recovery-related

124 Bladder Surgery for Benign Disease

1 · Surgical anatomy essentials

- Extraperitoneal hollow organ (mucosa, lamina propria, detrusor, perivesical fat); from urogenital sinus (endoderm).
- **Anterior**: space of Retzius, the usual access; transversalis fascia separates bladder from abdominal wall. **Posterior**: peritoneum, reflecting near the prostatorectal junction (men) or vesicouterine space (women); Denonvilliers fascia separates bladder/prostate from rectum. **Lateral**: pelvic sidewall, obturator space; vas or round ligament.
- **Medial umbilical ligaments** = obliterated umbilical arteries; **median umbilical ligament** = urachus (allantois remnant → patent urachus, sinus or cyst).
- Ureter crossings: **uterine artery** over ureter near internal os ("water under the bridge"); **vas** over ureter near seminal vesicle tip. Distal ureter runs medial to the medial umbilical ligament.
- Superior vesical (from umbilical) and inferior vesical (internal iliac; vaginal artery in women) arteries form posterolateral pedicles. Lymph mainly external iliac, also obturator, internal and common iliac.
- Nerves: sympathetic T10–L2, parasympathetic S2–S4; pudendal (Onuf's nucleus, S2–S4) to urethral sphincter.

> **KEY POINT**
> Three landmarks organise every benign bladder operation: Retzius for access, the peritoneal reflection for the posterior bladder, the distal ureter for safety — they decide whether surgery stays extraperitoneal.

> **KEY POINT**
> Common aftercare: watertight **two-layer 2-0 polyglactin** closure (mucosa, then seromuscular), leak test, large-bore Foley ≈ **1 week**, self-suction drain out at ≈ **48 h** if output low, **cystogram before catheter removal**. High drain output → drain creatinine above serum = urine leak.

2 · Bladder diverticula

- P2 P3 Bladder diverticulum / Hutch diverticulum — *definition (no muscularis propria), congenital vs acquired table, VCUG/cystoscopy/CT/urodynamics work-up, indications, tumour inside → cystectomy*
- Fibrous **pseudocapsule** gives the surgical plane. Extra causes: posterior urethral valves, prune-belly, iatrogenic (women after anti-incontinence surgery). Congenital (Hutch): still screen for voiding dysfunction; plan reimplantation.
- Cystoscopy: enter every sac; note neck size, distance from orifice, intravesical lobe, stricture or bladder neck contracture (treat before or with diverticulectomy); remove stones at surgery.
- Work-up extras: VCUG decides preoperative stent vs simultaneous reimplant; no contrast → ultrasound/non-contrast CT plus on-table retrograde studies; CT urogram for hydronephrosis (compression or VUR), ureteric displacement, stones; cytology in all; flow rate/urodynamics; prostate volume if treating BPH; **negative urine culture** before surgery.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection

Situation	Management
Large with residual, recurrent UTI, stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy ± reimplantation
Cancer in diverticulum	Partial or radical cystectomy — not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**; augmentation for neurogenic bladder. **Endoscopic outlet surgery first** — later irrigation would disrupt fresh suture lines.
- Localise the sac: Councill-tip catheter over a wire for selective filling, or flexible cystoscopic transillumination (best if multiple); ureteral catheter if near the ureter.
- Open surgery is extraperitoneal even for posterior sacs (avoids intraperitoneal leak) and remains gold standard; laparoscopic/robotic usually transperitoneal, reasonable in experienced hands.

> **ALGORITHM**
> Open intravesical: ASIS at table break → 18 Fr Foley, fill ≈200 mL → lower midline, split recti, incise transversalis fascia → sweep peritoneum off dome → vertical cystotomy, 2-0 stays → identify orifices (5 Fr ureteral catheter if concern) → excise sac → 20–22 Fr Foley, two-layer closure, leak test → optional suprapubic catheter → pelvic drain.

Open	Intravesical	Combined intra-/extravesical
Delivery	Allis clamp everts sac; or sac dissected from pseudocapsule from inside; mucosal incision ≈ **0.5 cm** round neck	Finger in sac pushes it out; extravesical neck incised circumferentially
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if incisions do not meet
Closure	Two layers 2-0 polyglactin, secure seromuscular layer (prevents recurrence); no drain in capsule	

Laparoscopic / robotic

- Lithotomy, steep Trendelenburg; cystoscopy first. Pneumoperitoneum **15 mm Hg**; assistant port contralateral to sac.
- Incise peritoneum over sac (drop bladder if large/multiple) → circumscribe neck → excise, bag; keep ureter in view. Staplers risk stones on staples.
- Transvesical variant (open bladder, invert sac) — cautious if ureter involved. Close in two layers 2-0 polyglactin; peritoneum may be closed; drain via trocar site.
- Sac at ureteral hiatus: amputate neck, close opening, new hiatus, double-J, reimplant.
- Alternatives for posterior sacs: extraperitoneal laparoscopy, pneumovesicoscopy, LESS, single-port robot. Drain out at 24–48 h.

Complications

- **Urinary extravasation commonest** (open = minimally invasive): high drain output, infected urinoma; transperitoneal → ileus, urinary ascites. Drain creatinine; CT cystography.

> **ALGORITHM**
> Asymptomatic leak → keep Foley and drain → symptomatic collection → percutaneous drain → cystogram before removal → persistent → nephrostomies vs exploration and closure.

- Unrecognised ureteric injury (AKI, flank pain) → imaging → retrograde stent; fails → nephrostomy, delayed repair.
- Others: UTI, fistula, bladder spasm, wound infection, bowel injury.

> **KEY POINT**
> Diverticulectomy is a dissection problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely — and fix the outlet first.

3 · Bladder stones, foreign bodies and fistulae

- Cause is **stasis**: BOO, neurogenic bladder, augmented/substituted bladders; foreign-body nidus (mostly iatrogenic: mesh, migrated IUD).
- Mostly asymptomatic; else haematuria, irritative voiding, recurrent UTI, retention.
- **Non-contrast CT** best for size/volume, prostate volume, hydronephrosis; cystoscopy only for haematuria, suspected cancer or prostate assessment.
- Treat stone **and** cause; the concomitant operation often picks the route (large prostate → transvesical prostatectomy + cystolithotomy; diverticulum → diverticulectomy; neurogenic → enterocystoplasty).

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation → urethral stricture
Percutaneous cystolithotripsy	Large stones	Nephroscope, short stay, no stricture	Contraindicated after prior surgery, diversion or augmentation (bowel injury)
Open cystolithotomy	Very large burden; with prostatectomy, diverticulectomy, augmentation	Historic gold standard	Incision, longer catheter
Laparoscopic/robotic	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technical demand

- **Percutaneous**: 1-cm incision two fingerbreadths above symphysis → needle under ultrasound/cystoscopic vision → dilate to **30 Fr** sheath → 24–26 Fr nephroscope; stones <1 cm grasped intact, larger fragmented. Suprapubic catheter clamped at 24–48 h.
- **Open**: Pfannenstiel or midline → vertical cystotomy → remove stones, inspect for tumour and intravesical lobe, do concomitant surgery → two-layer closure → drain. Prior surgery before percutaneous access: image to exclude interposed bowel.
- **Transvesical laparoscopy**: pneumovesicum **8–12 mm Hg**; avoids peritoneum; suits eroded mesh. Transperitoneal laparoscopic/robotic cystolithotomy allows simultaneous prostatectomy or diverticulectomy. Aftercare: Foley ≈1 week, cystogram.
- Refractory fistulae (colovesical, vesicovaginal) → partial cystectomy; refractory urinary fistula may justify simple cystectomy.

4 · Partial cystectomy for benign disease and urachal surgery

- Indications: leiomyoma, paraganglioma, endometriosis, urachal anomalies, fistulae (most benign tumours need only TUR).

- **Avoid with small capacity or severe dysfunctional voiding.** Exclude malignancy first (needs wide margins, lymphadenectomy). Lesions near the orifice need reimplantation.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; endovesical, intramural or extravesical	Transvaginal US, MRI	Small endovesical → TUR; large/intramural → wide local resection
Paraganglioma	Episodic hypertension; **hypotension/syncope on micturition or sex**; or painless haematuria	Plasma metanephrines; MIBG; yellow raised mucosa	**α-blockade before any instrumentation**; partial cystectomy over TUR (recurrence)
Endometriosis	Cyclical haematuria with menses	US, MRI (T1 bright); cystoscopy limited	Partial cystectomy (disease extravesical)
Urachal anomaly	Umbilical discharge, mucinuria, infected mass; malignant risk	US/CT; VCUG or sinogram; cystoscopy	Culture → antibiotics ± drainage; recurrent → excise urachus + dome cuff

- Dome/anterior → extraperitoneal; posterior → transperitoneal (mobilise ureter; develop pubocervical or Denonvilliers fascia for base lesions). No lymph node dissection. Localise by cystoscopy or ultrasound; **aspirate before opening to avoid spillage**.
- Full-thickness excision with ≈ **2-cm margin** if capacity allows; frozen-section margins; reimplant if orifice involved; 22 Fr Foley, two-layer closure, drain. Minimally invasive: flexible cystoscopic transillumination or laparoscopic ultrasound to outline the margin; ureteral catheter protects ureter.
- **Urachal excision**: whole tract with peritoneal wings plus small dome cuff; **no wide margin or frozen section**; umbilicus may be kept if benign. Open: periumbilical midline incision, circumscribe umbilicus, excise urachus down to a transmural dome cuff. Minimally invasive: camera 5 cm above and 2 cm left of umbilicus; urachus dissected last.
- Complications: extravasation commonest, then ureteric injury; reduced capacity — counsel.

> **KEY POINT**
> Leiomyoma, paraganglioma and endometriosis infiltrate and recur after incomplete excision, so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, margin). Urachal surgery is the exception: the whole tract matters, not a wide margin.

5 · Simple (supratrigonal) cystectomy

- Last resort for end-stage refractory dysfunction: radiation cystitis, neurogenic bladder, refractory incontinence, fistula, bladder neck contracture, refractory anastomotic stricture, IC/BPS. Aim: preserve renal function or quality of life.
- Prove the bladder is the source (non-vesical pain persists). IC/BPS success predictors: **small fibrotic bladder, small capacity under anaesthesia, Hunner lesions**.
- Diversion alone risks pyocystis, bleeding, pain, secondary cancer; many IC/BPS patients later need cystectomy.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed

Feature	Simple (benign)	Radical (cancer)
Lymphadenectomy	None	Performed
Bladder opened	Often — to find trigone and bladder neck	Avoided
Excision	Bivalve, amputate near trigone; stump oversewn, residual mucosa cauterised	En bloc

Steps: divide ureters → fill and bivalve bladder in midline → divide pedicles, amputate halves → posterior peritoneum covers stump → usually ileal conduit. Care mirrors radical cystectomy; robotic may reduce complications; drains 5–7 days (out when drain creatinine equals serum); complications common (wound infection, UTI, transfusion, ileus).

6 · Ureteral reimplantation in adults

- Indications: distal ureteral trauma, iatrogenic injury, **stricture** and delayed-presentation injury (commonest elective); with diverticulectomy or partial cystectomy. Primary VUR rarely.
- Acute: thermal or gunshot injury → debride even normal-looking ureter; partial injury → on-table retrograde pyelogram or methylene blue.
- Stricture causes: impacted stone, radiation, chronic inflammation. Atrophic kidney → renogram (decompress first). Stent in place → retrograde study before removal.
- Measure the gap with antegrade plus retrograde imaging. Exclude malignancy (cytology, ureteroscopic biopsy, frozen section). **Poorly functioning kidney → nephrectomy, not reimplant**. **Persistent extravasation despite adequate stenting** is an indication.

> **KEY POINT**
> Refluxing is acceptable in adults: VUR is rarely the indication, and refluxing anastomoses show no faster renal decline, more stenosis or more infection than tunnelled ones, while needing less ureter and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / mobilisation	More	Less — to posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical	Cephalad-medial to native hiatus	Submucosal tunnel; right-angle clamp draws ureter through
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over Kelly clamp	Clamp pulls ureter in

- Find ureter at the iliac bifurcation or beneath the obliterated umbilical artery; divide proximal to disease (no intramural dissection); frozen-section margin in stricture; mobilise

bladder off peritoneum (divide ipsilateral medial umbilical ligament ± urachus); test reach on the **decompressed** dome.

- **Lich-Gregoir**: small transmural cystotomy (neo-orifice) → serosa-detrusor incision toward bladder neck sparing mucosa → spatulated 4-0 anastomosis → stent → close detrusor over ureter with 2-0.
- **Politano-Leadbetter**: anterior cystotomy → ≈1.5-cm mucosal neo-orifice superior-medial to native → scissors raise submucosal tunnel → right-angle clamp punctures posterior wall, draws ureter through → anastomose, stent, close.
- Poorly vascularised ureter: non-transecting side-to-side refluxing anastomosis.
- Full-thickness ureteric sutures, mucosa plus some detrusor on bladder, knots outside lumen. Stent: **6 Fr × 26–28 cm double-J**.
- Preserve periureteric adventitia (no skeletonisation); extra ureteric mobilisation buys length but devascularises.
- Primary reimplant for defects up to **4 cm**; longer → psoas hitch/Boari; renal descent adds up to **5 cm**. More bladder reach: divide contralateral medial umbilical ligament, free peritoneum off dome, ligate contralateral pedicle.
- Enter **superior-medial, toward the bladder neck**; lateral sites kink on filling.
- Open preferred for trauma and remains gold standard; laparoscopic/robotic: less blood loss, shorter stay, equivalent patency.
- Aftercare: Foley 1 week with cystogram; stent **≈6 weeks**.
- **Leak** → drain urinoma, keep Foley and stent; after stent removal → re-stent; refractory → nephrostomy. **Late obstruction**, usually after stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site) → antegrade + retrograde imaging → endoscopic for short strictures, redo for high-grade. Others: UTI, fistula, spasm, stent colic.

> **KEY POINT**
> Reimplantation fails from ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other. The robot mainly eases intracorporeal suturing.

7 · Bridging distal ureteral defects

- P3 Boari flap — *reconstructive ladder, Boari dimensions (base ≥4 cm, apex ≥3 cm, ≤3:1, measured empty, reach up to 15 cm), psoas hitch nerves, contraindications, advancement flap*

Option	Reach	Rule
Primary reimplant	≤4 cm	Toward bladder neck
Psoas hitch	Gains ≈5 cm	Sutures superficial, parallel to fibres; reimplant before tying
Boari flap	Up to **15 cm**	Base ≥ **4 cm**, apex ≥ **3 cm**, length:base ≤ **3:1**

- Hitch + flap shortens the flap and preserves capacity. Hitch: oblique anterior cystotomy across about half the dome; 4–6 2-0 sutures into psoas minor tendon above the iliac bifurcation, avoiding **genitofemoral** (surface) and **femoral** (deep) nerves; reimplant on the superolateral dome.
- Bladder advancement flap (transverse incision one-third from dome to neck): reach comparable to Boari, simpler. Contraindication to hitch/flap: **small-capacity, poorly compliant bladder**.

- Hitch mobilisation: divide contralateral vas/round ligament ± contralateral superior vesical artery; refluxing reimplant if ureter short; 4-0 adventitial anchoring sutures.
- Boari: base on diseased side, anchored to psoas; tubularised in two layers. Still short → more ureteric dissection or renal descent.
- Minimally invasive: hitch to psoas minor superolateral to iliac vessels; refluxing extravesical reimplant; flap cut starting ≈2 cm from bladder neck; flap outlined with methylene-blue distension; 7 Fr stent.
- Complications: femoral/genitofemoral neurapraxia, iliac vessel injury, ischaemic stricture of narrow-based flap, reduced capacity with urgency/frequency. Beyond reach: transureteroureterostomy, ileal ureter, autotransplantation.

KEY POINT
The bladder is the donor tissue: hitch first, flap second — every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and bladder capacity.

8 · Enterocystoplasty (augmentation)

- Indication: neurogenic bladder with poor compliance **despite** CIC, anticholinergics, botulinum toxin and sacral neuromodulation — especially upper tract deterioration or incontinence with skin breakdown.
- Bowel lowers pressure only if **detubularised** (abolishes peristalsis).
- Work-up: video-urodynamics (detrusor leak point pressure predicts renal risk); serum chemistry, B12, fat-soluble vitamins; renal failure worsens acidosis; bowel history; **confirm ability to catheterise**. Metabolic effects → *Ch 133*.
- Add as needed: sling/AUS/bladder neck closure, catheterisable stoma, nonrefluxing reimplant for high-grade reflux.

Segment	Note
Ileum (commonest)	25 cm taken **≈20 cm** proximal to ileocaecal valve (spares B12/bile salts); acidosis risk
Ileocaecal	When a catheterisable channel is wanted
Sigmoid	Colonoscopy first
Stomach	Possible; higher incision

ALGORITHM
Isolate ileum, check reach to dome → transilluminate mesentery, staple, broad mesenteric base → side-to-side anastomosis, close window → **detubularise** on antimesenteric border, fold into U → **bivalve bladder widely** → wide-mouthed anastomosis → suprapubic tube ± Foley → leak test, drain.

- Narrow mouth → poorly emptying pseudodiverticulum.
- Early: extravasation (poor suturing, flap ischaemia — preserve arcade), UTI, bowel injury, fistula, intra-abdominal infection.
- **Perforation**: missed catheterisation; often masked by sensory loss → CT cystogram → **emergency exploration**.
- Stones commonest long-term complication. Neoplasia → surveillance endoscopy from **10 years**. Irrigation may reduce mucus. Routine metabolic/vitamin bloods. Long-term continence and satisfaction high, but secondary procedures often needed.

- Cannot catheterise urethra → continent ileocaecocystoplasty; unable to catheterise → incontinent ileovesicostomy.

KEY POINT
Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem: detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, bloods and endoscopic surveillance pay the price.

9 · Continent catheterisable channels

- P2 Principles of continence in continent diversion; Mitrofanoff principle — *submucosal tunnel flap valve, appendix first choice, Yang-Monti alternative, channel catheter 2–4 weeks, continence >95%, stomal stenosis commonest*
- Assess dexterity, habitus, neurology, bowel history (IBD, radiation, appendicectomy), renal function, video-urodynamics. Channel must reach stoma without tension, angulation or ischaemia.
- For adequate-capacity bladders (neuropathic, refractory stricture) after failed urethral CIC; augment if compliance poor. Always have a fallback if the appendix is absent or short.
- Monti uses 2–4 cm ileum; double Monti (diverticula), spiral Monti (more revisions). Vas, colon, stomach channels do poorly.

ALGORITHM
Minimally invasive appendicovesicostomy: detach appendix with caecal cuff → spatulate → ≈4-cm submucosal detrusor tunnel → anastomose with 4-0 polydioxanone → stoma right lower quadrant or umbilicus; bladder drained **2–4 weeks**; continence **>95%**.

KEY POINT
Use the appendix when present and long enough; reconfigure ileum when not. Continence comes from tunnel length through detrusor; missed catheterisation risks the bladder, not just the channel.

10 · Vesicovaginal fistula (VVF) †

- P3 P2 VVF: management options, early vs delayed repair — *causes, double-dye test, CT urography, catheter trial, timing, vaginal (Latzko, flap-split) vs abdominal (O'Conor) routes, interposition flaps*
- Outside source notes (†): leak usually 1–3 weeks after hysterectomy; coexisting ureteric injury about 10%; ureterovaginal fistula → stent first, then ureteroneocystostomy ± psoas hitch.

★ Putting it together

A · NMIBC: risk → treatment

AUA/SUO risk after complete TURBT.

Risk	Features	Adjuvant	Cystoscopy
Low	LG solitary Ta ≤3 cm; PUNLMP	Single post-op instillation	3 mo, then yearly
Intermediate	LG Ta recurring <1 yr/>3 cm/multifocal; LG T1; HG Ta ≤3 cm	6-week induction MMC/gemcitabine or BCG; maintain 1 yr	3–6-monthly 1–2 yr
High	HG T1; HG Ta >3 cm/multifocal/recurrent; CIS; variants; LVI; HG prostatic urethra	Re-TURBT (T1); BCG + **maintenance 3 yr** (SWOG); early RC if highest-risk	3–4-mo ×2 yr → 6-mo ×2 yr → yearly; upper tracts yearly ×2 yr
BCG-unresponsive	HG despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC; unfit: pembrolizumab (CIS), nadofaragene, gem–docetaxel	—

B · MIBC: cystectomy vs preservation → *Ch 131, 132*

	NAC + RC + PLND	Trimodality therapy
Who	Default if fit; NAC (gem–cis or ddMVAC) if cisplatin-eligible	Solitary T2, complete TURBT, **no CIS/hydronephrosis**, good function; or unfit
Cisplatin-ineligible	GFR <60, ECOG ≥2, grade ≥2 hearing loss/neuropathy, NYHA III–IV → local therapy	
After	Adjuvant nivolumab (CheckMate 274): ≥pT3/N+ post-NAC, or no NAC	Lifelong cystoscopy; salvage RC 10–30%

C · Choosing the diversion → *Ch 133–135*

	Ileal conduit	Orthotopic neobladder	Continent cutaneous
Best for	Frail, poor renal function, cannot catheterise, neurological	Default continent, either sex	Unusable urethra (margin tumour, stricture) or sphincter
Must have	Usable (non-irradiated) ileum	eGFR **>35–40**; intact sphincter; negative urethral frozen section; will catheterise	eGFR >35–40; dexterity; efferent limb (appendix, tapered ileum)
Main problems	Stoma problems, ureteroileal stricture	Nocturnal leakage; late retention (women); acidosis	Stomal stenosis, catheterisation difficulty, stones

D · Metabolic complications by segment

Segment	Disturbance	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis** (NH_4^+, Cl^- absorbed)	Bicarbonate, potassium citrate; chlorpromazine/nicotinic acid
Jejunum	**Hyponatraemic hypochloraemic hyperkalaemic acidosis**	NaCl, thiazide; avoid jejunum

Segment	Disturbance	Treatment
Stomach	**Hypochloraemic hypokalaemic alkalosis**; haematuria–dysuria	H2 blocker/PPI

Terminal ileum loss → low B12, bile-salt diarrhoea. Acidosis → bone buffering → osteomalacia. → *Ch 133*

E · Metastatic urothelial carcinoma

ALGORITHM

1. **Fitness**: cisplatin-, platinum-only or platinum-ineligible.
2. **First line**: EV + pembrolizumab (EV-302); else gem–cis (gem–carbo if cisplatin-ineligible).
3. **Non-progression on platinum**: avelumab maintenance (JAVELIN Bladder 100).
4. **Later**: pembrolizumab; erdafitinib (FGFR-altered); EV; sacituzumab govitecan. Platinum-ineligible: pembrolizumab first. → *Ch 131*